

Evidence Intelligence Engine

Product Scope of Work (Draft)

Purpose & Vision

The Evidence Intelligence Engine is an AI-powered investigative workspace for law enforcement agencies. It brings an investigation's evidence into one connected, searchable system and adds an AI reasoning layer on top of it — helping investigators retrieve information, explore relationships, reconstruct events, and generate reports through natural language, while every AI-generated finding stays traceable back to its supporting evidence.

The goal is for investigators to be able to interact with a case as though speaking to another investigator — asking questions, validating assumptions, and producing reports — without losing sight of the underlying evidence at any point.

Primary Users

- Investigation Officers
- Senior Investigation Officers
- Supervisors / Review Officers
- Intelligence Analysts
- Evidence Management Personnel
- Digital Forensics Teams

High-Level Workflow

Phase 1 — Case Creation

The investigator creates an investigation with core case details:

- FIR Number
- Case ID
- Crime Category
- Investigation Officer
- Police Station
- Incident Date
- Investigation Status

Along with, if available: Location, Initial Case Description, Victim Information, and Suspect Information.

Phase 2 — Evidence Ingestion

Evidence for a case will be ingested from multiple sources — documents, media, and structured records — through integration with existing databases and systems. The exact evidence types and formats will be defined in more detail as the system is scoped further.

Phase 3 — Automated Evidence Processing *(tentative)*

Once evidence is ingested, the platform is intended to automatically process it — for example OCR (at a later product stage), audio transcription (at a later product stage), metadata extraction, named-entity extraction, tagging, and duplicate detection — to make it searchable across the investigation. These processing steps are tentative and will be finalized after both teams get further into scoping and experimentation.

Modules

The modules below reflect the full long-term product vision, described at a high level. Not all of them are priorities for the initial Proof of Concept.

1. Investigation Workspace

A centralized dashboard for managing an investigation — case overview, evidence organization, status, and investigator notes — with a bilingual English / Urdu interface, including a language selection option.

2. AI Investigation Assistant

A natural-language interface investigators can query on a case — e.g. “summarize this investigation” or “what evidence supports the primary suspect?” — with source-cited answers, run when asked rather than continuously in the background.

3. Semantic Evidence Search

Search across a case's evidence using natural language rather than exact keywords, so relevant material surfaces even when the wording doesn't match exactly.

4. Case Summarization & Reporting

On-demand, AI-assisted summaries and draft reports — case overview, timelines, evidence, entities — that investigators can review and edit before export to PDF or Word.

5. Entity Intelligence

Structured profiles the platform identifies for key entities in a case (persons, vehicles, weapons, phone numbers, etc.), each carrying a confidence score — how certain the system is about that entity match or linkage — which investigators are expected to verify rather than take as final.

6. Timeline Reconstruction

An editable, chronological view of case events built from the available evidence, which investigators can adjust, annotate, and lock as verified.

7. Case & Pattern Analysis

On request, analyzes a case's evidence in different ways: surfacing relationships across evidence (shared people, vehicles, locations), flagging potential inconsistencies such as conflicting statements or timelines for investigator review, and looking for patterns or shared entities across multiple cases — plus aggregate

insights across the evidence repository, such as crime trends or hotspot identification. All of this is surfaced as things for investigators to look into, never as established fact.

8. Collaboration *(optional)*

Shared notes and comments between investigators on a case. Given the small number of users expected at this stage (roughly 1–5), this — and any related access/user-management overhead.

9. Audit & Chain of Custody

A log of who accessed or modified evidence and when, to preserve accountability for every action taken on a case.

10. AI Investigation Copilot

Beyond answering direct questions, the AI can proactively flag gaps or suggest next steps when an investigator is reviewing a case — again, at the investigator's request rather than running continuously in the background.

Platform Notes

- Bilingual English / Urdu interface, including AI-generated responses and reports
- On-premises deployment, to preserve data sovereignty
- Role-based access control, tied to the audit log in Module 9
- AI responses stay traceable back to their supporting evidence

Future Expansion Opportunities

The architecture should accommodate future capabilities without requiring significant redesign, including:

- CCTV integration
- Facial recognition integrations
- License plate recognition
- Biometric databases
- GIS mapping
- Crime prediction models
- Mobile investigator applications
- Court workflow integration
- Digital chain-of-custody systems
- External intelligence database integrations

Deliverable Objective

The outcome of this engagement is an AI-native Evidence Intelligence Engine that lets investigators search, analyze, connect, reason over, and report on multi-source evidence within one workspace — augmenting,

not replacing, human investigators, with every AI-generated finding kept transparent, reviewable, and traceable to its source evidence.

Note: This document reflects the full long-term product vision. The exact scope for the initial Proof of Concept will be discussed and finalized separately between both teams.